

hoBIT: A Profile-Aware Retrieval-Augmented Chatbot for University Academic Advising

Anonymous ACL submission

Abstract

In university academic advising, identical questions have different correct answers depending on a student’s department, admission cohort, and degree program. We present **hoBIT**, a profile-aware RAG chatbot that replaces a deployed RASA predecessor at a university college of informatics. It treats a schema-typed per-session profile as a first-class retrieval signal: *offline*, it tags the corpus with profile metadata; *online*, it acquires the profile from the student by *dual-asking* and injects it into retrieval as a *soft prefix* and a *hard filter*. On a 3,400-case, three-track evaluation we report three findings: (i) once the profile is present, *profile-blind* reranking and HyDE become redundant; (ii) the gain is a hard-filter / soft-prefix synergy, the soft prefix dominating; and (iii) the pipeline holds on open-weight models—more faithful on a judge-free metric—for low-cost on-premise deployment. This large profile-driven gain is headroom our benchmark exposes by design; the contribution is what realizes it and what it renders unnecessary. We release the system, a live demo, a screencast, and the evaluation suite.

1 Introduction

Retrieval-augmented generation (RAG) is increasingly tailored to specific domains (Gao et al., 2023b)—from medical education (Jang et al., 2025) to university student support (Tienes et al., 2025). Academic advising is one such domain: the *correct* answer to a recurring question—graduation requirements, required courses, double-major rules—is fixed by a small, *explicit* student profile, so a CS ’20 and a DS ’22 student asking the identical question need different curriculum tables (Figure 1).¹ At our college of informatics these questions burden departmental staff and a year-old FAQ chatbot that

¹CS, DS, and AI denote the Computer Science, Data Science, and Artificial Intelligence departments, respectively.

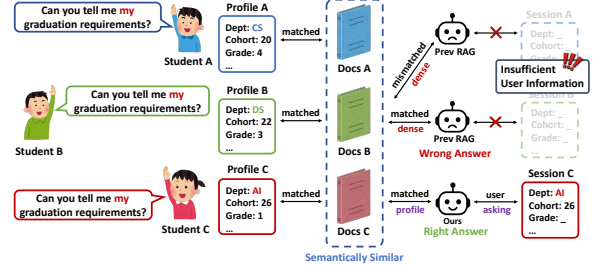


Figure 1: Same question, different correct answer per student. A profile-blind retriever (*Prev RAG*) returns the wrong cohort’s document or lacks the user information; hoBIT (*Ours*) acquires the needed profile fields via *dual-asking* and conditions retrieval on them.

cannot tailor answers to a student’s department, cohort, and major type.

This is a *profile-dependent* retrieval problem. Because curriculum documents are *semantically near-identical* across departments and cohorts, a profile-blind dense retriever cannot tell them apart and surfaces the wrong cohort’s table; the corpus already contains the right chunk (Lewis et al., 2020), so the bottleneck is *disambiguation by user-side structure*. The deployed RASA (Bocklisch et al., 2017) predecessor, built on rule and FAQ matching, fails exactly here—it cannot answer first-person, profile-dependent queries.

The disambiguating signal is a small structured profile—department, cohort, major type—which hoBIT treats as a *first-class retrieval signal*. Prior work personalizes retrieval differently: inferring free-form user personas from interaction history (Zerhoubi and Granitzer, 2024), learning user-specific preferences (Choi et al., 2025), or—in deployed university assistants—retrieving over shared, admin-curated content (Tienes et al., 2025). None conditions retrieval on an *explicit, schema-typed* profile acquired on demand, the regime academic advising requires.

To acquire it without friction, hoBIT is *session-based* rather than login-based—advising queries

are short and one-off, so accounts would add friction and a privacy liability. A TTL-bounded session profile is filled on demand by *dual-asking* (*proactively* when intent prediction flags a field, *reactively* when a cited chunk still requires one) and injected into hybrid dense–sparse retrieval as a *soft prefix* and a *hard filter*. Our central claim is not that personalization helps—on a profile-dependent benchmark that is expected—but that profile injection *reshapes what else matters*: it renders profile-blind reranking and query reformulation redundant, decomposes into a filter–prefix synergy, and holds with open-weight models for low-cost on-premise deployment.

We organize the evaluation around four research questions:

RQ1 Dominance. How large is the profile contribution relative to the pipeline’s other retrieval components?

RQ2 Subsumption. Does profile injection render second-stage reranking and query reformulation (HyDE) unnecessary?

RQ3 Mechanism. What drives the gain—hard payload filtering or soft query-prefix injection?

RQ4 Deployability. Does the pipeline hold with open-weight retrievers and generators, enabling low-cost, on-premise deployment?

Contributions. (1) hoBIT, a deployment-ready profile-aware advising chatbot that enriches its corpus with profile-based metadata offline and, online, acquires a per-session profile by *dual-asking* and injects it as a hard filter and a soft prefix. (2) A 3,400-case, three-track evaluation suite and ablation harness over open and proprietary models. (3) Evidence that profile injection subsumes profile-blind reranking and HyDE, decomposes into a filter–prefix synergy, and holds on open-weight models—the large profile-driven gain being headroom our benchmark exposes by design. A live demo and source code are available.²

2 The hoBIT System

hoBIT centers on a schema-typed *user profile*—department, major_type, grade, admission_year, and student_status—as a first-class retrieval signal (Figure 2). The system follows an offline–online design. Offline, the corpus is enriched with profile-based metadata.

Online, hoBIT acquires a session profile via dual-asking and injects it into a hybrid retriever as a hard metadata filter and a soft query prefix, focusing retrieval on profile-relevant evidence. Incoming queries are first routed by intent, with only retrieval queries entering the profile-aware pipeline, while the remaining intents are handled directly. Frequently asked administrative questions are answered from an administrator-maintained FAQ store, avoiding unnecessary LLM generation. Retrieved evidence is then passed to the generator, which produces grounded responses with sentence-level source citations. Additional implementation details are deferred to Appendix A.

Offline: profile-based semantic enrichment.

Every chunk is LLM-tagged (OpenAI, 2024a) with two kinds of profile label: *value tags*—a department tag and an admission_year range—that the hard filter uses to exclude other cohorts, and a required_profile list naming which profile fields the chunk’s answer depends on, which drives reactive dual-asking. Content is indexed in lifecycle-split static/dynamic collections.

Online: acquiring the profile (dual-asking).

Once a query is routed as a *retrieval* intent, hoBIT predicts *which* profile fields it needs—its required_profile (e.g., department and admission_year for “my graduation requirements?”)—and checks them against the session. Missing fields are requested by *dual-asking*, at two points: **proactively**, before retrieval, for fields the prediction flags; and **reactively**, when a *cited* chunk’s required_profile names a field the session still lacks (e.g., major_type). In either case hoBIT withholds the answer and asks for the missing field; the response is produced on the next turn, once the student supplies it. Values are cached in the Redis session—no account or password—and reused silently, so each field is asked at most once.

Online: injecting the profile (hard & soft).

hoBIT then injects the acquired profile at the two points where retrieval can use it. As a **soft prefix**, the profile is prepended to the query (“my graduation requirements?” → “Dept: CS, Cohort: 20; ...”), steering dense and lexical matching toward the student’s cohort. As a **hard filter**, the same profile constrains the candidate set through the chunk payload labels, dropping other-cohort chunks while keeping profile-agnostic content. The

²<https://xxx.example.com>

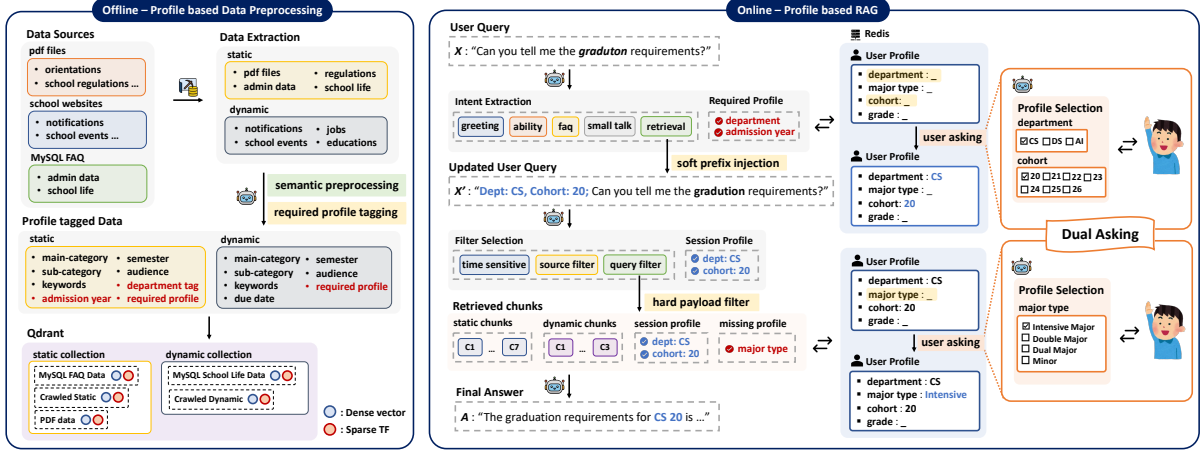


Figure 2: hoBIT centers on a schema-typed *user profile*. **Offline (left)**: every chunk is LLM-tagged with the profile labels it presupposes—department, admission_year, and its *required_profile*—and indexed in Qdrant. **Online (right)**: hoBIT predicts the profile fields a query needs, acquires missing ones by *dual-asking* into a Redis session, and injects the profile as a soft query prefix and a hard payload filter before generating a source-tracked answer; a still-missing field (e.g., major type) triggers a reactive ask.

two are complementary—the soft prefix shapes *what ranks high*, the hard filter controls *what is eligible*. The generator then emits an answer with per-sentence $[S_i]$ citations.

3 Experimental Setup

Datasets (three tracks, 3,400 unique queries). All three tracks run over one index crawled from our college (curriculum PDFs, department pages, an operational FAQ), with categories curated from the index and validated against one year of the deployed predecessor’s real query logs—for validation only, preserving privacy and reproducibility (§Appendix B). **(i) Intent routing** (1,600): five-class query classification (greeting, ability, FAQ, smalltalk, and retrieval); its retrieval class reuses the 1,200 queries of (iii), so the three tracks span 3,400 unique queries. **(ii) Profile-grounded QA** (1,800), our main track: deterministic curriculum questions with a known correct source, constructed so the answer is cohort-determined, isolating the profile-dependent regime. **(iii) Open-ended advising** (1,200): administrative and school-life queries without ground truth, over categories disjoint from (ii). The profile is held out of every query and set through the session, so the ablation measures profile-aware retrieval’s *pure* contribution.

Ablation systems (RQ1). We compare five incremental configurations of the same pipeline (Table 1), each adding one component over a dense retriever: Dense, Lifecycle, Hybrid, Filtered, and

Configuration	Component added	Retrieval		Profile	
		S/D	BM25	Filt.	Prof.
Dense	dense retriever (baseline)	–	–	–	–
+Lifecycle	static/dynamic index split	✓	–	–	–
+Hybrid	BM25 / RRF fusion	✓	✓	–	–
+Filtered	metadata payload filter	✓	✓	✓	–
hoBIT	profile injection (soft+hard)	✓	✓	✓	✓

Table 1: Ablation configurations (RQ1): each row adds one component—*S/D* static/dynamic split, *BM25/RRF* fusion, *Filt.* payload filter, *Prof.* profile injection. **hoBIT** is the full system; results in Table 2.

hoBIT. This design isolates the contribution of each retrieval component, culminating in profile-aware retrieval with profile injection.

Model pools (RQ2–RQ4). **RQ2** adds two profile-blind boosters—a cross-encoder reranker (BAAI, 2024) and HyDE query reformulation (Gao et al., 2023a). **RQ3** only toggles the profile-injection mode (soft prefix, hard filter, or both). **RQ4** substitutes open models on each stage, holding the other fixed: open embedders BGE-M3 (Chen et al., 2024) and Qwen3-Embedding (Qwen Team, 2025a) against proprietary text-embedding-3-small (OpenAI, 2024b), and open generators Qwen3-8B (Qwen Team, 2025b) (multilingual) and EXAONE-3.5-7.8B (LG AI Research, 2024) (Korean-specialized) against gpt-4o-mini (OpenAI, 2024a). The full retriever sweep is reported in Table 4.

System	CtxP	CtxR	Kw.	Src.M	KO-F	PASS
Dense	0.665	0.851	0.599	0.030	0.901	2.1%
+Lifecycle	0.710	0.889	0.605	0.024	0.899	1.9%
+Hybrid	0.749	0.880	0.587	0.068	0.905	6.2%
+Filtered	0.746	0.885	0.591	0.070	0.903	6.8%
hoBIT	0.718	0.894	0.701	0.585	0.888	47.4%

Table 2: **RQ1: deployed component ablation** (1,800 cases; configs in Table 1). The profile step drives nearly all of the gain. Src.M=graded Source Match; metric definitions in §Appendix E. Best **bold**.

Metrics and evaluation modes. We evaluate in two modes. (a) RQ1 runs the deployed system (dual *static* + *dynamic* collections), scored by a graded *Source Match*, *Keyword Match*, and *deepeval* (Confident AI, 2024) LLM-judge metrics. (b) RQ2–RQ3 use a controlled single-static-collection setup with binary rank metrics (MRR, R@1/5/10); the RQ4 generator comparison instead reuses the quality metrics of (a). The two modes are consistent but not numerically identical. Metric definitions, judge configuration, and the Source Match grading are in §Appendix E.

4 Results

hoBIT is strong across all three tracks: it far outperforms the best non-profile system on Profile-grounded QA (Table 2), reaches 96.0% intent-routing accuracy, and scores 0.796 Overall on the (ground-truth-free) open-ended track. We dissect *why* through RQ1–RQ4 on Profile-grounded QA.

4.1 RQ1: Profile Injection Dominates

By design these answers are cohort-determined, so the magnitudes quantify the *headroom* the benchmark exposes; the real questions are how to realize it (§4.3) and which standard components it renders redundant (§4.2).

Profile injection dominates. Adding the profile (Filtered → hoBIT) is by far the largest jump on every headline metric (Source Match 0.070 → 0.585, PASS 6.8% → 47.4%). The retrieval-only components leave the profile-dependent answer near-unreachable: Hybrid nearly triples Source Match (0.024 → 0.068), but on a negligible base still far from hoBIT. Two minor judge-metric regressions reflect the greater specificity of profile-aware answers.

Filtering without profile is a no-op. Hybrid → Filtered turns on the payload-filter path without profile data to populate it, contributing

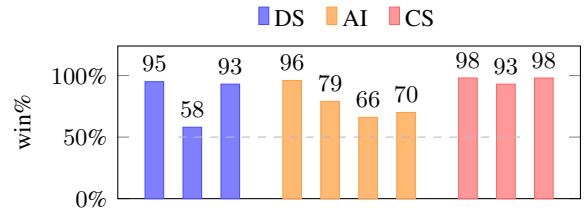


Figure 3: Blind pairwise study ($n=48$): **hoBIT** wins all 10 profile-dependent pairs (**85.3%** non-tie, $p < 0.001$). Bars: per-pair win%, colored by department; dashed: 50%.

Configuration	MRR	R@1	R@5	R@10	s/q
No profile	0.038	0.009	0.069	0.152	–
+reranker	0.089	0.027	0.165	0.322	0.55
+HyDE	0.037	0.016	0.055	0.138	3.38
+soft prefix	0.276	0.151	0.464	0.614	–
+hard filter	0.189	0.112	0.280	0.433	–
hoBIT (soft+hard)	0.538	0.462	0.652	0.711	0.38
+reranker	0.364	0.252	0.496	0.629	0.58
+HyDE	0.431	0.326	0.569	0.683	3.31

Table 3: **RQ2–RQ3: controlled study** (single static collection; rank metrics over top-10). Upper rows build from a no-profile baseline: reranking lifts it (still far below hoBIT) while HyDE does not, and *soft/hard* (RQ3) help far more, together forming **hoBIT**. Lower rows add profile-blind reranking/HyDE *on top of* hoBIT and fall below it. Best **bold**; *s/q* latency—shown for hoBIT and the reranker/HyDE stages, omitted (–) for profile-only variants (embedding-API-bound, near-constant); HyDE rows include document generation.

nothing—filter and profile are a single mechanism (§4.3).

Hybrid earns its keep at depth. Its top-rank effect is small without the profile, but hybridization lifts Recall@50 from 0.81 to 0.92 (full sweep in Table 4), widening the candidate pool the profile filter then narrows.

Human preference. In a blind pairwise study ($n=48$, A/B counterbalanced, the *same* profile given to both systems), hoBIT beat the Dense baseline on *all* 10 profile-dependent pairs ($p < 0.001$; Figure 3).

4.2 RQ2: Profile over Reranking and HyDE

We compare profile injection against two common RAG enhancements: reranking and query reformulation (HyDE) (Table 3).

Reranking is redundant. A cross-encoder reranker helps a profile-blind baseline but *hurts* on top of the profile: the profile already ranks the correct cohort curriculum first, and a profile-blind

Retriever	MRR	R@1	R@5	R@10	R@50
BM25 (Kiwi)	0.425	0.318	0.594	0.706	0.926
openai_small	0.515	0.438	0.627	0.677	0.805
+BM25	0.537	0.459	0.652	0.711	0.916
BGE-M3	0.605	0.552	0.681	0.691	0.757
+BM25	0.593	0.525	0.692	0.720	0.906
Qwen3-Emb	0.601	0.544	0.685	0.704	0.784
+BM25	0.588	0.516	0.693	0.731	0.906

Generator	Src.M	Kw.	KO-F	Ovr.	tok	s/q
gpt-4o-mini	0.619	0.665	0.927	0.737	171	3.5
Qwen3-8B	0.653	0.694	0.941	0.763	252	7.1
EXAONE-7.8B	0.667	0.674	0.923	0.755	253	5.8

Table 4: **RQ4: open-model deployability** (1,800 cases). *Top*: retriever sweep—each embedder alone and as a shaded +BM25 (Kiwi (Lee, 2024)) hybrid, **bold** = larger of the two per column; openai_small+BM25 is deployed hoBIT. *Bottom*: generator sweep (shared retrieval), best per column **bold**. *tok* mean output tokens, *s/q* latency.

reranker reorders it downward by surface similarity.

HyDE is redundant. A hypothetical-document query cannot encode the student’s department and cohort, so it does not even improve the no-profile baseline, while a *trivial* profile prefix far outperforms it; stacked on the profile it again does not help, yet it adds a full LLM call (~ 3 s/query, Table 3) for no gain.

These profile-blind boosters help *at most* a weak baseline and become redundant—or harmful—once the profile is present. A *profile-aware* second stage might still help; we claim only that the standard, profile-blind versions do not.

4.3 RQ3: What Drives the Profile Gain

The profile is injected as a *hard* payload filter and a *soft* query prefix (§2); we decompose the two (Table 3).

Soft prefix drives recall; top-1 needs both. The soft prefix alone helps recall more than the hard filter—prepending department/cohort tokens pulls the correct curriculum into the top-10—so the filter is not the whole story. But top-1 is super-additive: only together do the two rank the gold document first (the filter removes wrong-cohort distractors, the prefix ranks the correct table up)—a *filter-prefix synergy* that explains the no-op of §4.1.

4.4 RQ4: Open Models for Deployment

Open retrievers match or exceed proprietary. Across seven retrievers, open embeddings match or

beat the proprietary baseline (Table 4, top); a fully open retriever sacrifices nothing.

Open generators are at least as good. With retrieval fixed (Table 4, bottom), both open models match or exceed gpt-4o-mini; on the *deterministic, judge-free* Source Match they significantly beat it (paired bootstrap, CI excludes zero). Since the judge is itself gpt-4o-mini, any self-preference bias works *against* the open models, making these estimates conservative. They are also more verbose (~ 250 vs 171 output tokens), which—with single-slice batch-1 serving—explains their higher latency.

Efficiency. Search cost is equal across embedders; the gap is the embedding step (remote API vs. local GPU), so open embedding is also faster (appendix Table 6).

Deployment. Our pilot uses gpt-4o-mini: it receives *no* personally identifying data (only cohort attributes) and is called on demand, so a proprietary API is a reasonable default today. Still, the results show an open retriever and generator enable a *fully on-premise* pipeline at no measurable quality cost—and better faithfulness—which we treat as future work.

5 Demonstration

We walk through hoBIT’s online loop (Figure 2) on the live web UI (Figure 4): a *cold start* that acquires a profile by *dual-asking*, and a *warm session* that reuses it—so each field is asked at most once.

UC1: Cold-start acquisition. A student asks “*what are my required major courses?*” with no profile stored. A *proactive* prompt collects department/admission_year (CS/2020); a cited chunk then also needs major_type—required courses differ for basic vs. intensive major—so a *reactive* prompt follows (*intensive major*). hoBIT caches the profile and answers the correct cohort curriculum with [Si] citations.

UC2: Warm-session reuse. The student next asks “*what are my graduation requirements?*”. The fields are already cached, so *no* prompt appears—the stored profile silently drives the hard filter and soft prefix, and the cited answer is immediate, until the session TTL expires.

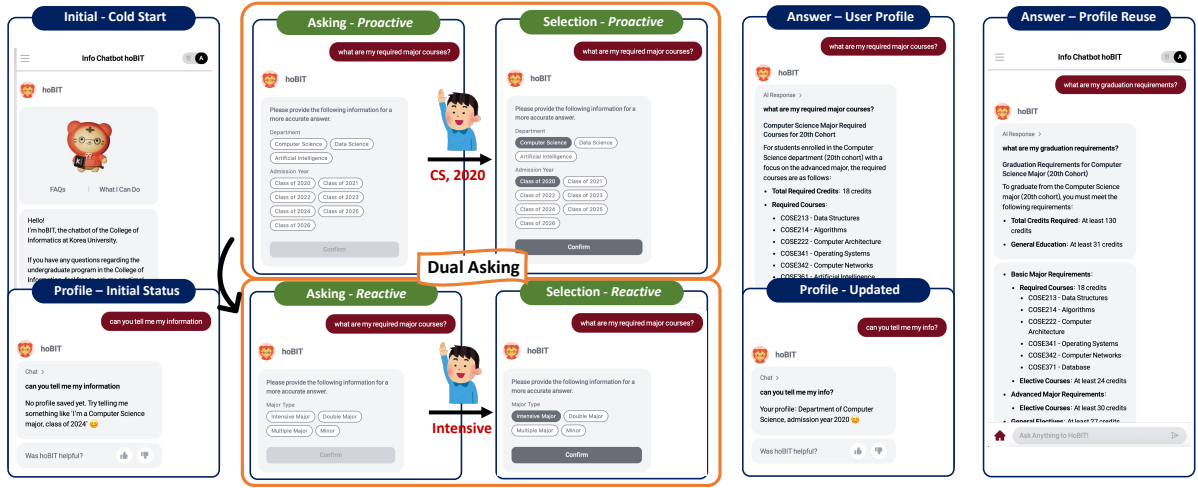


Figure 4: The hoBIT web interface: *dual-asking* acquires the profile at cold start (*proactive* then *reactive* ask-and-select), after which hoBIT answers with source citations and reuses the stored profile.

6 Related Work

Retrieval-augmented generation. RAG fuses dense (Karpukhin et al., 2020) and lexical (Robertson and Zaragoza, 2009) retrieval (Cormack et al., 2009), with reranking (Nogueira and Cho, 2019) and query reformulation (Gao et al., 2023a); §4.2 shows these profile-blind boosters become redundant under profile injection.

Personalization in LLMs. Recent work personalizes the *model/preferences* (Choi et al., 2025; Thonet et al., 2025; Yunusov et al., 2025) or *retrieval* from history (Qin et al., 2025; Li et al., 2025). hoBIT instead injects an explicit *schematyped* profile as a hard filter and soft prefix, needing no per-user learning; Qin et al. (2025)’s value-over-similarity finding mirrors our RQ2 result that profile-blind reranking can *hurt*.

Academic-advising chatbots. Closest is Marcel (Trienes et al., 2025), which returns the *same* answer to every student. hoBIT conditions on a per-student profile—identical questions yield student-specific answers—with dual-asking acquisition and source-tracked generation; to our knowledge the first deployed advising system to quantify schematyped profile injection against reranking, hybrid, and query-side components.

7 Conclusion

hoBIT is a deployed profile-aware RAG chatbot for academic advising. On a 3,400-case evaluation, profile injection *reshapes what else matters*: it renders profile-blind reranking and HyDE redundant, rests on a filter–prefix synergy, and holds on

open-weight models—more faithful on a judge-free metric—for low-cost on-premise deployment. The large profile-driven gain is headroom our benchmark exposes by design; the value is in what realizes it and what it renders unnecessary. We release the system, a live demo, and the evaluation suite.

Limitations

(1) **Scope.** A single college; schema and pipeline are domain-general, but the curriculum and 3,400-case suite are institution-specific, so external validity is untested. (2) **Two evaluation modes.** RQ1 (dual collections, graded Source Match) and RQ2–RQ4 (single static, binary R@K) are consistent but not identical, so we report each separately. (3) **Latency.** Open-model latencies (single MIG slice, batch 1) are not comparable to the proprietary API. (4) **Deployment gap.** The pilot serves gpt-4o-mini; the open-model backend is validated offline but not yet live. (5) **Profile correctness.** hoBIT trusts self-declared fields, unverified against a student record.

Acknowledgments

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Appendix A Runtime Pipeline

The deployed system runs as four stages over two endpoints (`/classify`, `/chat`), consulting and (dual-)asking for a schema-typed Redis profile and citing sources throughout (Figure 5).

Stage 1 (`/classify`). `gpt-4o-mini` returns the intent, a normalized query, keywords, and the `required_profile` (forced to include `department/admission_year` for curriculum queries). If a required field is absent from both query and session, the backend *blocks before* `/chat`—no retrieval or generation runs.

Stage 2 (`/chat`). The profile enters retrieval through three channels: a Qdrant payload filter (`is_null OR match`, keeping profile-agnostic chunks), a query prefix, and a system-prompt hint. Dense (`text-embedding-3-small`) and sparse (BM25/Kiwi) top-50 lists are fused by RRF ($k=60$) to top-10, with a *static / dynamic* time gate. An explicit department/cohort in the query overrides the profile (a fallback, not a hard constraint).

Stage 3. After generation, only the *cited* chunks’ `required_profile` tags are aggregated; a still-missing field discards the draft and issues a reactive prompt.

Stage 4. Each sentence carries a `[Si]` citation (else the model abstains); these tags drive the UI source links, the Stage 3 parse, and the Source Match metric (§Appendix E).

Appendix B Dataset Construction

Index. One index of 906 chunks (curriculum PDFs, orientation booklets, department pages, an operational MySQL FAQ), segmented and LLM-tagged as in §2.

Real-usage anchors. From 3,058 query rows of the RASA predecessor, Korean-only / dedup / length filters keep 859 *anchors*; top-3 retrieval keeps the 797 in a valid index category. Anchors only *validate* category coverage—never generation prompts—avoiding student PII and enabling reproduction elsewhere.

Curation. Categories are curated *bottom-up* from the index and checked *post-hoc* against anchors: negligibly-supported cells are dropped (anchor-spec match $\approx 51\%$, equal to the index’s own coverage, i.e., unbiased).

Track II (Profile-grounded QA). The 60 profiles are 15 department-cohort pairs (grouped by curriculum-revision years, e.g. CS revised 2020/2021/2024/2025) $\times$ 4 grades, each crossed with 10 profile-dependent categories and 3 query styles (*formal*, *personal*, *verification*) via LLM paraphrase. The profile is never written into the query text, isolating profile-aware retrieval.

Tracks III and I. 12 administrative/school-life categories, disjoint from Track II, with queries synthesized from their reference chunks (no ground truth; LLM-judge scored). Track I reuses the 1,200 Track III queries as its *retrieval* class plus 400 non-retrieval queries (100 each greeting/ability/FAQ/smalltalk).

Appendix C Per-Category Source Match (Profile-grounded QA)

Table 5 breaks the RQ1 profile gain down by category. The lift is largest on cohort-partitioned categories (general vs. intensive major +0.90, major-required +0.75) and smallest on double-major credit (+0.14), whose rules cut across cohorts.

Academic category	+Filtered	hoBIT	lift
General vs. intensive major	0.03	0.93	+0.90
Major-required courses	0.03	0.78	+0.75
General-education required	0.12	0.78	+0.66
Department-signature electives	0.22	0.84	+0.62
Graduation credit decomposition	0.07	0.61	+0.54
Double-major credit (<i>worst</i>)	0.04	0.18	+0.14

Table 5: Profile lift is largest on cohort-partitioned categories.

Appendix D Latency Decomposition

Table 6 splits retrieval latency into embedding and Qdrant search: search is near-constant, so the gap is the embedding step (proprietary API vs. local GPU).

Retriever	embed (s)	search (s)	total
BM25 (Kiwi)	0.003	0.048	0.052
openai_small	0.264	0.064	0.328
+BM25	0.257	0.102	0.358
BGE-M3	0.056	0.061	0.116
+BM25	0.055	0.093	0.148
Qwen3-Emb	0.060	0.060	0.120
+BM25	0.061	0.093	0.154

Table 6: Retrieval latency (controlled study). Open embedders run on a single MIG GPU slice.

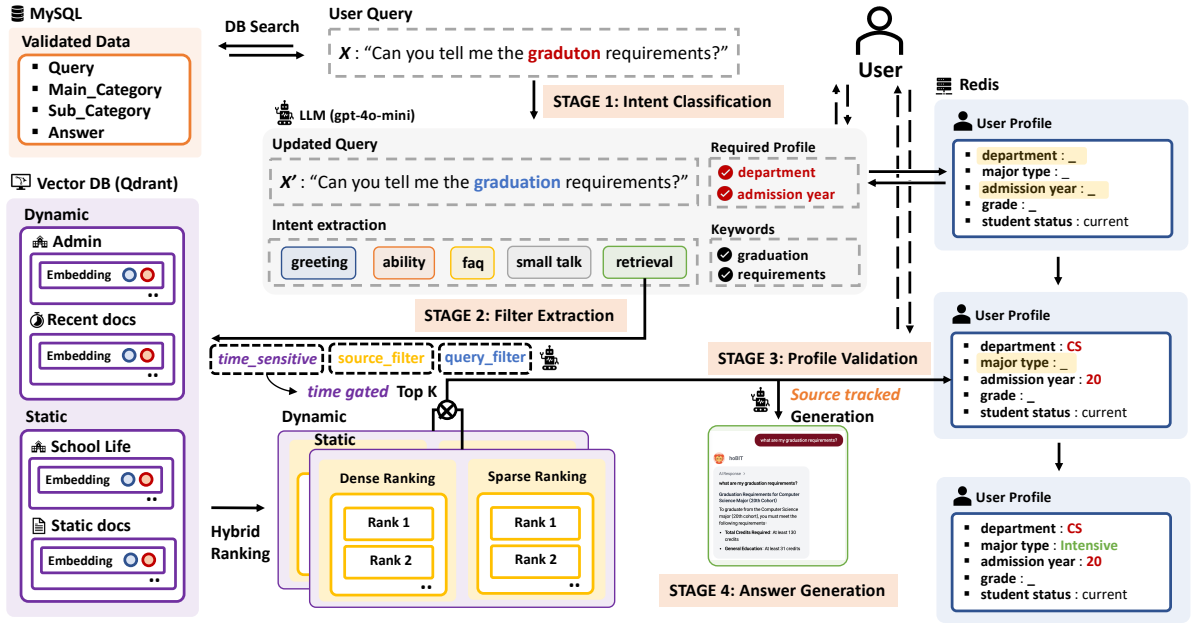


Figure 5: The four-stage runtime: (1) intent classification with a proactive profile check, (2) filter extraction and hybrid dense+sparse retrieval, (3) reactive validation on the cited chunks, and (4) source-tracked generation.

Appendix E Metric Definitions

Deterministic. *Keyword Match*: fraction of expected content keywords appearing in the response. *Source Match* (RQ1): graded attribution over the deployed dual collections—gold source cited/retrieved/missing = 1/0.5/0—so it credits citation, not just retrieval. *Recall@K* (RQ2–RQ4): binary hit@K over one global ranking of a single static collection (pure retrieval, hence higher absolute values). **LLM-judge (gpt-4o-mini via deepeval)**. *Contextual Precision/Recall*: ordering and coverage of retrieved chunks. *Korean Faithfulness* (GEval): every claim grounded in context, Korean synonym-aware (a claim-free answer scores 1.0). *Answer Completeness* and *Top-3 Precision* (open-ended track) reward honest “no information” and top-3 relevance. “Overall PASS” = passing all metrics at threshold 0.5.

Appendix F HyDE Example

For “what is the credit difference between my basic and intensive major?”, HyDE generates a fluent but profile-agnostic passage (e.g., “the basic major requires at least 36 credits, the intensive major at least 24...”) that hallucinates numbers and ignores the student’s department and cohort—exactly the disambiguation the task needs, so it cannot retrieve the correct curriculum (§4.2).

GenAI Usage Disclosure

We use gpt-4o-mini inside hoBIT as the answer generator and during evaluation as the LLM judge (via deepeval) for Contextual Precision, Contextual Recall, Korean Faithfulness, Top-3 Retrieval Precision, and Answer Completeness; it also generates the HyDE hypothetical documents used in the RQ2 comparison. The open-weight generators evaluated in RQ4 are Qwen3-8B and EXAONE-3.5-7.8B-Instruct. All deterministic metrics (Keyword Match, Recall@10, MRR, intent-classification accuracy) are computed without LLM judgement. Full prompt templates, per-metric configurations, and the demonstration video are hosted on the project homepage. The authors additionally used general-purpose chat assistants during paper drafting (organization and copy-editing) and as a programming aid for portions of the implementation; all technical claims, system-design decisions, experimental analyses, and final code review originate with and are the responsibility of the authors.